How Deep Is Your Data? Estimating Recursive Generation Depth via Statistical Fingerprints

Anonymous ACL submission

Abstract

As language models increasingly train on web text containing their own outputs, a recursive model→data→model loop arises, yet no method estimates how many such iterations a text has undergone. We formalize *recursive generation depth estimation* as ordinal classification, constructing a 180K-passage benchmark spanning three model families, three decoding strategies, and four depths under controlled full-replacement recursive chains at the 1–1.5B scale. A Random Forest over statistical features dominated by the perplexity tail family achieves d1–d3 balanced accuracy of 59–82% (four-class: 69–86%) across nine model–strategy cells; convergence of RF, XGBoost, and LightGBM confirms that discriminative signal resides in feature design, not classifier choice. Cross-model transfer reaches 66.4%, cross-domain arXiv 74.1%, and cross-strategy transfer 59.1%; the gap reveals that decoding strategies induce structurally distinct degradation trajectories, formalized via the *Effective Discriminable Depth* (EDD) metric ($\text{EDD}_{0.70}{=}4$ for all three families). Binary detectors and membership inference baselines (30–35%) fall far below our pipeline; ordinal depth estimation requires distributional statistics inaccessible from surface text.

1 Introduction

The text on which large language models are trained increasingly contains their own outputs. As synthetic content proliferates across the web, a recursive loop emerges: models trained on web corpora ingest text generated by earlier models, and their outputs feed into the next generation's training data (Shumailov et al., 2024; Alemohammad et al., 2024). This model→data→model cycle is no longer hypothetical; it is already occurring in the wild. Yet whether the recursive generation depth of a text can be estimated from its statistical properties remains unstudied, a question that matters because first-generation synthetic text may be benign or even beneficial for training, while deeply recursive text drives the distributional collapse that degrades model quality (Dohmatob et al., 2024b). We study this question under *controlled full-replacement chains*, where each generation fully replaces the prior training set, isolating the depth variable. Real-world contamination typically involves partial mixing (accumulation), where synthetic and human data co-exist at unknown proportions; the full-replacement setting provides an upper bound on depth signal strength, and we discuss implications for accumulation settings in §6.

Existing tools address adjacent but distinct problems. Binary detectors such as Binoculars (Hans et al., 2024), Fast-DetectGPT (Bao et al., 2024), and DivEye (Basani and Chen, 2026) classify text as human-written or machine-generated, collapsing all depths $d \geq 1$ into a single category. Watermarking (Kirchenbauer et al., 2023) requires proactive embedding and cannot be applied retroactively. The model collapse literature (Shumailov et al., 2024; Dohmatob et al., 2024b) assumes depth is known a priori; Drayson et al. (2025) filter synthetic data via binary detection without distinguishing depths. No existing method estimates the ordinal recursive generation depth of a text passage from its content alone.

We find that recursive generation leaves quantifiable statistical fingerprints in text, and these fingerprints are **consistent in direction but scale-dependent in magnitude** across the three model families and three decoding strategies studied. Specifically, the perplexity tail feature family (`p90_ppl`, `p75_ppl`, `kurtosis_ppl`) and surprisal variance emerge as dominant depth indicators, with group-level dominance robust across all 9 model–strategy configurations and three importance measures. Instruction-tuned variants exhibit monotonically decreasing AUC (regularization), whereas base models can show non-monotonic tra-

jectories; instruction tuning acts as a regularizer on the depth signal rather than a source of degradation.

Following the tradition of benchmark-driven empirical studies in NLP, our contribution centers on task formalization and systematic empirical characterization rather than algorithmic novelty; the convergence of multiple classifiers (RF, XGBoost, LightGBM within 0.4 pp; §5.1) confirms that discriminative signal resides in feature design, not classifier architecture. We formalize recursive generation depth estimation as ordinal classification and introduce a lightweight pipeline: compute 19 reference-model statistical features (effective dimensionality ≈4–5 due to multicollinearity; §4.2) and classify with a Random Forest into $K+1$ depth classes. To characterize how recursive generation reshapes these features, we construct the *Depth Degradation Atlas*: a 3×3 grid of model families $\times$ decoding strategies spanning 4 depths (180K samples). The atlas reveals strategy-dependent degradation trajectories and we introduce the *Effective Discriminable Depth* (EDD) metric to quantify how far into the recursion chain adjacent depths remain distinguishable.

Our contributions are:

- **Task formalization and benchmark.** We define recursive generation depth estimation as ordinal classification, introduce the EDD metric, and construct a 180K-sample benchmark (3 model families $\times$ 3 decoding strategies $\times$ 4 depths).
- **Depth Degradation Atlas.** The perplexity tail feature family dominates depth discrimination across all 9 cells; $\text{EDD}_{0.70}{=}4$ for all three families, validated by 5-class (d0–d4) extension (71–80%); at $T{=}0.75$, Pythia achieves $p{=}0.001$ with graded confidence across families.
- **Generalization and cross-strategy analysis.** Cross-model LOO 66.4% (d1–d3: 55.4%), cross-domain arXiv 74.1%, single-cell prompted 69.5%, preliminary cross-scale 7–14B (54–70%, attenuating with scale). We identify a structural barrier in cross-strategy transfer (59.1%, d1–d3: 45.9%): decoding strategies induce qualitatively distinct degradation trajectories, with d1/d2 boundary inversion under strategy mismatch. Surface-text controls (zero-shot LLMs 30–36%) and binary detection/membership inference baselines (DetectGPT, Min-K%: 30–33%) confirm the signal resides in distributional statistics inaccessible from surface text.

2 Related Work

Model Collapse Theory. Recursive training on model outputs erodes distributional tails (Shumailov et al., 2024), an effect formalized as exponential tail decay via scaling laws (Dohmatob et al., 2024b; Seddik et al., 2024) and extended to autophagous loops in the MAD framework (Alemohammad et al., 2024). Subsequent work refines the conditions under which collapse occurs: Bertrand et al. (2024) identify convergence to a fixed point under mixed real-synthetic retraining (providing theoretical grounding for our EDD ceiling, §3.5), Shi et al. (2025) reframe collapse as a generalization-to-memorization transition, and Dohmatob et al. (2025) extend guarantees to broader model classes. Mitigation (Gerstgrasser et al., 2024; Drayson et al., 2025; Yi et al., 2026; Falahati et al., 2026; Wenzek et al., 2020), data scaling under constraint (Muennighoff et al., 2023), non-monotonic dynamics (Bakshi and Chakraborty, 2026), convergence analysis (Marchi et al., 2024; Wang et al., 2026; Keisha et al., 2025; Dohmatob et al., 2024a), and diversity decline (Guo et al., 2024b; Briesch et al., 2024; Padmakumar and He, 2024; Gu et al., 2026; Schaffelder and Gatt, 2026) are active areas. Schaeffer et al. (2025) argue that "model collapse" conflates multiple phenomena and call for operationally precise metrics; we respond by operationalizing tail-erosion predictions into a deployable ordinal classifier that reveals strategy-dependent trajectories and non-monotonic 7B behavior that the theory does not predict. All assume depth is known a priori; we address the inverse problem of estimating depth from text alone.

AI Text Detection. The detection literature (Wu et al., 2025) spans statistical methods (DetectGPT (Mitchell et al., 2023), Fast-DetectGPT (Bao et al., 2024), Binoculars (Hans et al., 2024), DivEye (Basani and Chen, 2026), BiScope (Guo et al., 2024a), GLTR (Gehrmann et al., 2019), Ghostbuster (Verma et al., 2024), DNA-GPT (Yang et al., 2024)), geometric approaches based on intrinsic dimensionality (Tulchinskii et al., 2023), and watermarking (Kirchenbauer et al., 2023), with known limits from paraphrasing (Sadasivan et al., 2023), distribution shift (Pudasaini et al., 2026), and generator barriers (Pu et al., 2026). M4-SemEval (Wang et al., 2024), RAID (Dugan et al., 2024), and MAGE (Li et al., 2024) benchmark detection across domains, models, and adversarial conditions. Bevendorff et al. (2025) distinguish authorship at-

tribution (which model?) from verification (human or AI?); depth estimation introduces a third axis: which recursive iteration. All these methods perform binary classification, discarding ordinal depth information; our approach outputs ordinal depth labels, complementing rather than replacing binary detection.

Membership Inference and Data Provenance. Min-K% Prob (Shi et al., 2024) detects pretraining membership via low-probability token statistics, related to our perplexity features but limited to binary classification; Datamodels (Ilyas et al., 2022) trace predictions to influential training examples; cryptographically undetectable watermarks (Christ et al., 2024) require generator cooperation. Depth estimation occupies a distinct niche: ordinal, retroactive, and cooperation-free.

Adjacent Tasks. EditLens (Thai et al., 2026) estimates editing extent but does not model recursive chains; Uchendu et al. (2020) classify *which* model generated text; Mohamed et al. (2025) characterize *how* text degrades under iterative generation without classifying which iteration a passage comes from. Our task performs the inverse: ordinal depth estimation from text alone.

Ordinal Classification. CORAL (Cao et al., 2020) enforces rank consistency via ordered binary tasks; CLOC (Pitawela et al., 2025) extends contrastive learning with multi-margin ordinal losses; Kasa et al. (2024) benchmark ordinal techniques for NLP; Janitza et al. (2016) show RF implicitly respects ordinal structure when inter-class distances are monotonic. We adopt CORAL and CLOC as baselines (§5).

Decoding Strategy Effects. Decoding strategies shape text statistics in ways that interact with depth signals (Holtzman et al., 2020; Finlayson et al., 2024; Garcés Arias et al., 2025; Wiher et al., 2022; Basu et al., 2021). Our atlas (§4) shows these strategies induce qualitatively different degradation trajectories under recursive generation.

3 Method

3.1 Problem Definition

Let f_θ denote a language model and $\mathcal{D}_0$ a corpus of human-written seed text. We define a *recursive generation chain* of depth K as the sequence $\mathcal{D}_0, \mathcal{D}_1, \ldots, \mathcal{D}_K$, where each $\mathcal{D}_k$ ($k \geq 1$) is produced by finetuning f_θ on $\mathcal{D}_{k-1}$ and sampling from the resulting model. Depth $d{=}0$ denotes human-written text; depth $d{=}k$ ($k \geq 1$) denotes text generated by a model trained on depth-$(k{-}1)$ data.

Given a text passage x, the task is to estimate its recursive generation depth $d \in \{0, 1, \ldots, K\}$. This is an ordinal classification problem: depths carry a natural ordering, and $d{=}1$ is structurally closer to $d{=}0$ than $d{=}3$ is, since each generation step compounds distributional drift.

3.2 Multi-Generational Benchmark Construction

We construct a benchmark spanning three axes: model family, decoding strategy, and generation depth.

Model families. We select three 1–1.5B-parameter models with diverse pretraining corpora and architectures: Pythia-1.4B (EleutherAI), OLMo-1B (AI2), and GPT-2 XL (OpenAI, 1.5B). Restricting to a narrow parameter range isolates the effect of architecture and training data from scale.

Decoding strategies. Each model is paired with three decoding strategies: nucleus sampling ($p{=}0.95$) (Holtzman et al., 2020), temperature sampling ($\tau{=}0.9$), and top-k sampling ($k{=}50$). These span the dominant modes of open-ended generation and induce qualitatively different entropy profiles. Our findings on feature importance and discriminability are limited to this grid; transfer to untested models, scales, or strategies is not established.

Depth and scale. The primary benchmark covers depths d0–d3 across all $3 \times 3 = 9$ model–strategy cells, with 5,000 samples per depth per cell (180K samples total). For depth extension experiments (§4), we extend one pipeline (Pythia-1.4B, nucleus) to d0–d5.

Generation pipeline. Starting from a Wikipedia subset ($\mathcal{D}_0$), each cycle: (1) LoRA finetuning (Hu et al., 2022) on $\mathcal{D}_{k-1}$ ($r{=}16$, $\alpha{=}32$, lr $2{\times}10^{-4}$, effective batch 16, 3 epochs, dropout 0.05; fresh adapter from the base model at each depth; full protocol in Appendix A); (2) sampling 5,000 passages (max 512 tokens, BOS-seeded) under the specified strategy; (3) $\mathcal{D}_k$ feeds the next cycle. Each depth is produced by a model trained on prior-depth data, not by sampling from the unmodified base model. BOS-seeded generation may amplify depth signal; prompted-generation validation (§5.3) confirms signal persistence (-13.8 pp). No per-passage lineage

Table 1: The 19 statistical features comprising $\mathbf{f}(x)$, grouped by category. All perplexity and surprisal features are computed from the reference model (Pythia-1.4B).

Category	Features	n
Perplexity statistics	mean, var, skewness, kurtosis, p10, p25, p50, p75, p90	9
Surprisal statistics	mean, var, entropy	3
Lexical diversity	type–token ratio, hapax ratio, bigram entropy, trigram entropy	4
Repetition	repeated 2/3/4-gram fractions	3

exists across depths; cross-validation splits do not introduce leakage.

Reference model. All statistical features (§3.3) use Pythia-1.4B as fixed reference model. This overlaps with one generator (Pythia), conferring a 4–11 pp self-reference advantage on that pipeline (Appendix B.13); Pythia results in Table 2 are marked † accordingly. Alternative references yield 67–79%, all exceeding 2.5× chance (Limitations).

3.3 Statistical Feature Design

We extract a feature vector $\mathbf{f}(x)$ comprising 19 candidate features from the reference-model perplexity distribution over each passage x; ablation (§4.2) shows the top 5 achieve 97.6% of full-set performance, and variance inflation analysis reveals VIF>10 for 14–16 features; effective independent dimensionality is ≈4–5. The features fall into four categories (Table 1).

Perplexity statistics (9 features) capture the per-token perplexity distribution via moments and quantiles (p10–p90); tail quantiles are especially informative as recursive finetuning erodes the high-perplexity tail (§4). *Surprisal statistics* (3 features): mean, variance, and entropy of $-\log p(\text{token})$. *Lexical diversity* (4 features): type–token ratio, hapax ratio, bigram/trigram entropy. *Repetition* (3 features): repeated n-gram fractions for $n \in \{2, 3, 4\}$.

3.4 Effective Discriminable Depth

Aggregate multi-class accuracy obscures whether discriminability degrades at specific depth boundaries. We propose the *Effective Discriminable Depth* (EDD) as an operational measure of how far into a recursive chain adjacent depths remain distinguishable.

Pairwise discriminability. Let $\mathcal{D} = \{0, 1, \ldots, K\}$ denote the depth set and $T \in (0.5, 1.0]$ a discrimination threshold. For each adjacent pair $(d, d{+}1)$, we define the pairwise discriminability $\mathrm{AUC}(d, d{+}1)$ as the area under the ROC curve of a dedicated binary Random Forest classifier (500 trees, $\lfloor\sqrt{19}\rfloor$ candidate features per split) trained on class-balanced data from $\mathbf{f}(x)$ (§3.3) to distinguish depth-d from depth-$(d{+}1)$ samples, evaluated via 5-fold stratified cross-validation. Each pair uses an independent classifier to avoid decision boundary dilution from multi-class settings.

Definition 1 (Effective Discriminable Depth)**.** The *Effective Discriminable Depth* at threshold T is:

$$\mathrm{EDD}_T = \max\{d^*{+}1 : \forall\, d' \le d^*,\ \mathrm{AUC}(d', d'{+}1) \ge T\} \quad (1)$$

where $d^* \in \{0, \ldots, K{-}1\}$. If $\mathrm{AUC}(0, 1) < T$, then $\mathrm{EDD}_T = 0$.

EDD_T reports the longest *contiguous* discriminable prefix: the deepest d such that every pair from $(0, 1)$ through $(d{-}1, d)$ exceeds T. All 12 AUC profiles in our benchmark decay monotonically, empirically validating the monotonicity assumption; the sole non-monotonic case (an instruction-tuned model outside the primary atlas) yields EDD=0, rendering the distinction between contiguous and non-contiguous discriminability moot for our configurations. We report T=0.70 as primary threshold, corresponding to adequate discrimination in ROC analysis (8/9 cells achieve EDD ≥ 3); sensitivity at $T \in \{0.65, 0.70, 0.75, 0.80\}$ in §4.

Classifiers. Our primary classifier is a Random Forest (200 trees, unconstrained depth, 5-fold CV).[1] The RF produces ordinally consistent predictions: 98.2% of errors fall on adjacent depths ($|\Delta| \le 1$), QWK = 0.89; CORAL ordinal regression matches within 0.6 pp ($p > 0.05$; §5). Comparisons against CORAL (Cao et al., 2020), DivEye (Basani and Chen, 2026), and other baselines appear in §5.

3.5 Theoretical Grounding

Recursive generation contracts the support of the learned distribution, with low-probability tokens lost first (Shumailov et al., 2024): tail mass

[1] EDD binary classifiers use 500 trees; the main 4-class RF uses 200 trees.

$P_k(\text{ppl} > t)$ decreases monotonically in depth k. This tail erosion predicts the perplexity tail feature family as a depth marker, confirmed by its dominance across all 9 cells (§4.2). Surprisal variance captures complementary distributional narrowing: as the support contracts, per-token surprise becomes more predictable.

EDD upper bound. Tail mass at depth k scales as α^k for a contraction factor $\alpha \in (0, 1)$ (Dohmatob et al., 2024b); EDD reaches its upper bound when adjacent-depth feature distance falls below classifier resolution. Our empirical AUC-excess contraction (ratio 0.80 at d1v2→d2v3, 0.69 at d3v4→d4v5; §4.4) is consistent with geometric decay approaching a fixed point (Bertrand et al., 2024). 5-class extensions to d4 (71–80%, chance = 20%; Appendix B.17) confirm depth 4 discriminability; mean perplexity saturates before tail statistics, placing a physical ceiling on EDD.

Scale-dependent attenuation. Larger models approximate the training distribution more faithfully per step (α closer to 1), so EDD should decrease with scale. Qwen2.5-14B confirms this: d0-vs-d1 AUC drops to 0.643 (vs. >0.99 at 1–1.5B), while d1–d3 AUCs remain above 0.70 (§5.4).

Strategy-dependent convergence. The contraction rate α varies with decoding strategy. Temperature scaling ($\tau < 1$) sharpens the logit distribution uniformly and accelerates convergence to the mode; temperature depth-d text should therefore resemble nucleus depth-$(d{+}1)$ text in tail statistics. GPT-2 XL confirms this: temperature d1 median `p90_ppl` (92.2) matches the nucleus d2 median (92.3, IQR 75.8–112.5). Top-k truncation removes the low-probability tokens that carry depth signal, predicting weaker discriminability; top-k cells yield the lowest accuracy (68.9–75.8%) across the atlas (§4.5).

4 The Depth Degradation Atlas

We present the Depth Degradation Atlas: a systematic empirical map of how statistical fingerprints evolve across recursive generation depths, models, and decoding strategies. The atlas spans $3 \times 3 = 9$ model–strategy cells evaluated at depths $d \in \{0, 1, 2, 3\}$, with depth extensions to $d{=}5$ for cross-model EDD validation (§4.4).

Table 2: Four-class RF accuracy (%) with bootstrap 95% CI (±half-width; $B{=}1{,}000$). RF with 200 trees, evaluated via 5-fold stratified CV (seed=42). Parenthesized: d1–d3 balanced accuracy from the same 4-class RF (chance = 25% / 33%). Depth-0 recall exceeds 97% in all cells. Within-distribution ordinal quality: MAE = 0.23, QWK = 0.89 ($\text{QWK}_{\text{d1–d3}} = 0.74$); 98.2% of errors fall on adjacent depths ($|\Delta| \leq 1$). †Self-reference scoring (Pythia-1.4B is both generator and reference model); 4–11 pp advantage over cross-model references (Appendix B.13).

	Nucleus	Temp. τ=0.9	Top-k
Pythia-1.4B†	78.7±0.6 (71.8)	76.8±0.6 (69.3)	68.9±0.6 (58.9)
OLMo-1B	78.9±0.6 (72.7)	82.3±0.5 (76.6)	74.4±0.6 (66.2)
GPT-2 XL	**84.3**±0.5 (79.6)	**85.9**±0.4 (81.5)	**75.8**±0.6 (68.1)

4.1 In-Distribution Performance

Table 2 reports four-class Random Forest accuracy (5-fold CV, 200 trees) across all nine cells. Accuracy ranges from 68.9% (Pythia, top-k) to 85.9% (GPT-2 XL, temperature) four-class (d1–d3 balanced: 58.9–81.5%), all exceeding the 25%/33% chance baselines. Two axis effects emerge. GPT-2 XL achieves the highest accuracy under every strategy (75.8–85.9%), followed by OLMo (74.4–82.3%) and Pythia (68.9–78.7%). Top-k sampling yields the lowest accuracy in every model. The mean within-model strategy gap is 9.3 pp, comparable to the 7.2 pp mean within-strategy model gap; decoding strategy and model family contribute roughly equally to performance variance.

Per-class recall. Depth-0 recall exceeds 97% in all 9 cells. Among generated depths, d2 consistently yields the lowest recall (44–73%), followed by d3 (61–80%) and d1 (72–91%). The d2 trough reflects its position at the center of the depth gradient, where features overlap with both d1 and d3, consistent with d2-vs-d3 as the EDD bottleneck (§4.4). Full per-class recall appears in Appendix B.4.

4.2 Feature Consistency Across the Grid

Despite the 17.1 pp accuracy spread across cells, the perplexity tail statistics family (`p90_ppl`, `p75_ppl`, `kurtosis_ppl`) and surprisal variance (`var_surprisal`) dominate depth discrimination in every setting. Within this family, `p90_ppl` ranks top-2 by Gini importance in all 9 cells and `var_surprisal` in 8/9 under SHAP (7/9 Gini; Appendix B.2), though individual rankings should be interpreted cautiously given high

Table 3: Top-5 features by Gini importance. **Bold**: top-2. `p90_ppl` top-2 in 9/9; `var_surprisal` 8/9 under SHAP (7/9 under Gini). Individual rankings are approximate due to multicollinearity (VIF>10 for 14–16/19 features).

Cell	Top-5 features (Gini importance, ranked)
Pythia, nuc.	**var_surp (.17)**, **p90_ppl (.15)**, m_ppl (.11), var_ppl (.08), p75 (.07)
Pythia, τ	**var_surp (.17)**, **p90_ppl (.14)**, m_ppl (.11), p75 (.07), m_surp (.07)
Pythia, top-k	**var_surp (.16)**, **p90_ppl (.12)**, m_ppl (.11), var_ppl (.07), m_surp (.06)
GPT-2 XL, nuc.	**var_surp (.21)**, **p90_ppl (.21)**, p75 (.10), m_surp (.07), m_ppl (.06)
GPT-2 XL, τ	**p90_ppl (.25)**, **p75 (.15)**, m_surp (.13), var_surp (.12), p50 (.05)
GPT-2 XL, top-k	**p90_ppl (.24)**, **p75 (.12)**, var_surp (.10), m_surp (.09), ttr (.04)
OLMo, nuc.	**p90_ppl (.18)**, **var_surp (.15)**, p75 (.11), m_surp (.07), ttr (.06)
OLMo, τ	**p90_ppl (.21)**, **var_surp (.16)**, p75 (.12), m_surp (.09), ttr (.05)
OLMo, top-k	**p90_ppl (.20)**, **var_surp (.17)**, p75 (.10), m_surp (.07), m_ppl (.05)

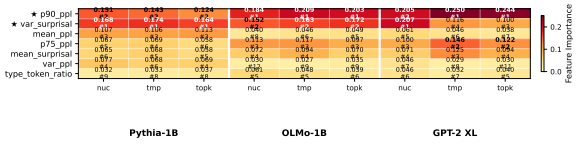


Figure 1: Atlas feature importance heatmap. Rows: 19 features grouped by category (§3.3); columns: 9 model–strategy cells. Color intensity encodes RF Gini importance. The perplexity-tail and surprisal-variance blocks maintain high importance across all cells.

multicollinearity (VIF>10 for 14–16/19 features).[2] Table 3 shows the full per-cell ranking.

Cross-model feature-rank Spearman correlation averages $\bar{\rho} = 0.868$ (range $[0.619, 0.954]$): models agree on which features carry depth information even when they differ in absolute accuracy. Among strategies, top-k yields the highest cross-model rank correlation ($\rho = 0.811$): the deterministic vocabulary truncation imposes a more uniform feature landscape across architectures, at the cost of weaker overall signal (§4.5).

Feature ablation confirms this concentration: the top-5 features achieve 97.6% of full accuracy (76.5% vs. 78.4%), with perplexity statistics alone reaching 76.6% (vs. surprisal 73.2%, other categories $\leq$47%).

The feature hierarchy shifts with depth. At shallow boundaries (d1 vs. d2), `var_surprisal` dominates importance; at deeper boundaries (d3 vs. d4, d4 vs. d5), tail perplexity statistics (`p90_ppl`, `kurtosis_ppl`) take over. This transition aligns with model-collapse dynamics (Shumailov et al., 2024): early recursion narrows the surprisal distribution (captured by its variance), while later steps erode the perplexity tail (captured by upper quantiles and kurtosis).

[2]The group-level dominance of perplexity tail statistics is robust across importance measures (Gini, SHAP, permutation; Kendall's W=0.90, p<0.005).

4.3 Monotonicity and Inter-Depth Distances

Depth features exhibit strong monotonic trends: perplexity tail statistics decrease with depth (Spearman $\rho = -1.0$ across cell-level medians) while repetition features increase (Figure 4). Inter-depth distances are non-uniform: d0-vs-d1 AUC exceeds 0.99 in all 9 cells, whereas d2-vs-d3 ranges from 0.69 to 0.89.

4.4 EDD Analysis

We extend the benchmark to depth d=5 for all three models under nucleus sampling. Each model's depth-3 LoRA checkpoint is used to finetune and generate depth-4 data, and the process repeats for depth 5; feature extraction uses the same reference model (Pythia-1.4B). We train dedicated binary RF classifiers (500 trees) for each adjacent pair, as defined in Eq. 1.

AUC decay and cross-model convergence. For Pythia-1.4B (nucleus), AUC decays monotonically from 0.999 (d0v1) to 0.683 (d4v5); AUC-excess contraction ratios (0.80 at d1v2→d2v3, 0.69 at d3v4→d4v5) show convergence toward indistinguishability accelerates with depth (Figure 2). The d3-vs-d4 AUC clusters near T=0.75 for all three models (Pythia 0.784, OLMo 0.765, GPT-2 XL 0.750), while d4-vs-d5 converges to a narrow band (0.681–0.684, range 0.003), suggesting a model-independent depth ceiling. Bootstrap testing (H_0: AUC $\leq$ 0.75) yields p = 0.001 (Pythia), 0.091 (OLMo), 0.505 (GPT-2 XL).

EDD threshold sensitivity. Across the 9-cell benchmark (depths 0–3), all cells achieve EDD=3 at T=0.65; at T=0.70, 8/9 cells; at T=0.75, 7/9; at T=0.80, 5/9. Top-k configurations lose EDD=3 first (Pythia top-k at T=0.70, OLMo top-k at T=0.75); d2-vs-d3 is the consistent bottleneck.

4.5 Strategy-Dependent Trajectories

The three strategies induce qualitatively distinct degradation trajectories. Temperature (τ=0.9) yields the highest accuracy in 2/3 models (GPT-2 XL 85.9%, OLMo 82.3%) because uniform logit scaling accelerates convergence to an attractor (Shumailov et al., 2024), amplifying the fingerprint but creating a boundary reversal risk under cross-strategy transfer (§5). Top-k (k=50) produces the lowest accuracy (68.9–75.8%) because hard vocabulary truncation removes depth-discriminative low-probability tokens. The strategy effect (9.3 pp range) exceeds the model effect

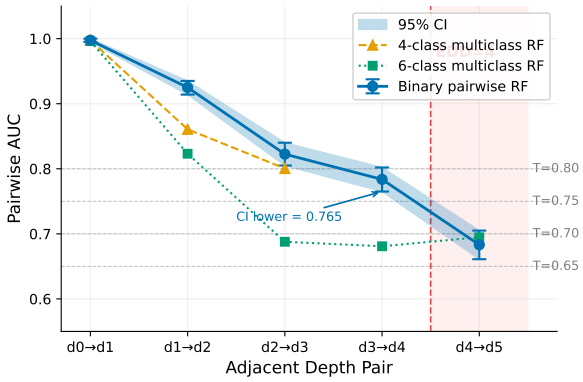


Figure 2: Binary pairwise AUC decay curves for three models (nucleus sampling, depths 0–5). Horizontal dashed lines mark $T \in \{0.65, 0.70, 0.75, 0.80\}$. All three curves converge below 0.70 at d4-vs-d5, while d3-vs-d4 AUC clusters in the range 0.750–0.784. Shaded bands: bootstrap 95% CIs.

(7.2 pp); deployment-time knowledge of the decoding strategy matters at least as much as the generator model.

5 Generalization and Robustness

We test whether atlas accuracy (68.9–85.9% four-class; d1–d3: 58.9–81.5%) reflects genuine statistical regularities across eight generalization axes (five below, three in Appendix).

5.1 Classifier Comparison and Surface-Text Ablation

Table 4 compares ordinal depth estimation methods across all 9 atlas cells. RF, XGBoost, and LightGBM on the same 19 features converge within 0.4 pp per cell (a feature-determined ceiling); on Pythia nucleus, CORAL (Cao et al., 2020) (78.1%) and DivEye (Basani and Chen, 2026) (78.2%) match RF within 0.6 pp ($p > 0.05$).

Binary detectors and membership inference. Four binary detection/membership inference methods, calibrated for ordinal depth, all fall 43–48 pp below RF (Table 4). Min-K% Prob (Shi et al., 2024) (33.2%) outperforms DetectGPT (Mitchell et al., 2023) (30.7%), but CORAL ordinal calibration does not close the gap (d1–d3: 40.4% vs. RF 71.6%); the bottleneck is input signal, not calibration (Appendix B.25). Adding Min-K% as a 20th RF feature yields only +0.48 pp (78.4%→78.9%); the 19 statistical features already capture the dominant ordinal signal.

Surface-text ablation. Single-feature regression on mean perplexity performs at chance

Table 4: Ordinal depth estimation: method comparison (mean across 9 atlas cells; DetectGPT: 8 of 9, Pythia nucleus excluded due to scoring instability). Chance = 25% (4-class) / 33.3% (d1–d3). Min-K%: K=20, CORAL calibration. DetectGPT: gpt2-large scoring, t5-large perturbation (n=5). 5-fold CV; SD $\leq$ 0.9 pp for feature-based methods.

Method	4-Cl Acc (%)	d1–d3 BalAcc (%)
RF 19-feature (ours)	**78.4**	**71.6**
Zero-shot LLM (best)	36.2	—
Min-K% Prob (CORAL)	33.2	40.4
DetectGPT (perturbation)	30.7	24.3
Random	25.0	33.3

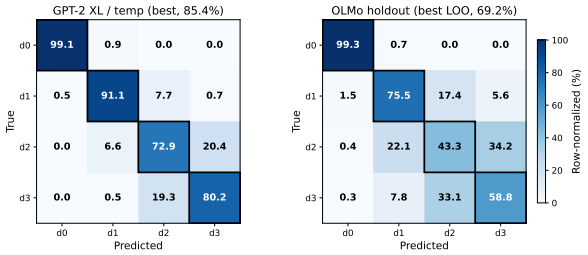


Figure 3: Confusion matrices for the best in-distribution setting (GPT-2 XL, temp τ=0.9, 85.9%) and the best cross-model leave-one-out transfer (OLMo holdout, 69.2%). d0 recall exceeds 99% in both; errors spread across the d1–d3 range under transfer.

(25.1%); zero-shot LLMs (DeepSeek V3.2, 36.2%; Qwen2.5-7B-Instruct, 30.4%) also fail (43 pp below the best RF cell; Appendix B.6); depth signal resides in distributional statistics inaccessible from surface text.

Convergence confirms the signal resides in the features: three classifiers match within 0.8 pp (Appendix B.20), perplexity features alone recover 66.0%, and seed replication across three families shows $\Delta \leq 3.2$ pp (Appendix B.16). Figure 3 shows d0 near-perfectly identified (recall >99%); errors concentrate at d2/d3 in-distribution and spread to d1–d3 under transfer.

5.2 Cross-Model Transfer

We perform leave-one-model-family-out evaluation: train on two families (nucleus only), test on the held-out third (Table 5).

Mean transfer accuracy is 66.4% four-class (d1–d3: 55.4%; 2.7× chance); d0 recall (>98%) inflates the four-class headline, and d2 recall drops to 26.8% in the worst case (Pythia holdout).

5.3 Cross-Strategy Transfer

Table 6 reports leave-one-strategy-out transfer.

Table 5: Cross-model transfer (leave-one-out, nucleus only). Chance = 25% / 33% (d1–d3). d1–d3 bal. acc: Pythia 49.9%, OLMo 59.2%, GPT-2 XL 57.2% (mean 55.4%). Ordinal quality: MAE = 0.39, QWK = 0.81 ($QWK_{d1–d3} = 0.53$); 94.8% of errors at $|\Delta| \leq 1$.

Holdout	Acc (%)	d0 Rec.	d1 Rec.	d2 Rec.	d3 Rec.
Pythia	62.4	100.0	49.3	26.8	73.6
OLMo	**69.2**	99.3	75.5	43.3	58.8
GPT-2 XL	67.6	98.5	74.7	52.6	44.5
Mean	66.4	99.3	66.5	40.9	59.0

Table 6: Cross-strategy transfer (leave-one-out). Chance = 25% / 33% (d1–d3). d0 recall ≥89% inflates headline; d1–d3 mean: 45.9%.

Holdout	Pythia Acc (%)	OLMo Acc (%)	GPT-2 XL Acc (%)
nucleus	**69.4**	**65.4**	**59.9**
temp (τ=0.9)	62.8	58.2	56.1
top-k (k=50)	51.5	54.9	53.6

Cross-strategy transfer yields 59.1% four-class (d1–d3: 45.9%, chance = 33%), 7 pp below cross-model transfer. This gap is itself an empirical finding: different decoding strategies induce structurally distinct degradation trajectories in feature space, not merely noisier versions of the same trajectory. Training on nucleus + top-k inverts the d1/d2 boundary under temperature (d1v2 AUC < 0.5), because temperature scaling accelerates convergence (§3.5). A two-stage pipeline (strategy identification + depth classification) recovers 11.5 pp, with a 31.5 pp d1–d3 gap between correct and wrong routing confirming feature-space misalignment (Appendix B.27).

Prompted generation. Replicating the GPT-2 XL nucleus pipeline with topic-seeded generation yields 69.5% (95% CI $[68.9, 70.2]$, -13.8 pp vs. BOS-seeded, $2.8\times$ chance), with monotonic AUC decay and identical top features (Appendix B.26): prompt conditioning attenuates but does not eliminate the depth signal.

5.4 Preliminary Cross-Scale Observations

Depth signal persists at larger scales: 7–8B models achieve 60–70% and 14B (Qwen2.5-14B) reaches 54.1% (d1–d3: 57.4%, chance = 33%), though confounded by five factors (quantization, sample size, single strategy, vocabulary mismatch, self-reference scoring; Appendix B.14). Cross-model chains (84.75%) and cross-domain arXiv (74.1%) confirm robustness at the primary 1–1.5B scale (Appendix B.14).

6 Discussion and Conclusion

This paper formalizes recursive generation depth estimation as ordinal classification under controlled full-replacement chains. The perplexity tail feature family dominates depth discrimination across all nine cells, with $EDD_{0.70}$=4 for all three families (confirmed by 5-class extension, 71–80%; Appendix B.17); cross-model transfer reaches 66.4% (d1–d3: 55.4%), cross-domain arXiv 74.1%, while binary detectors and membership inference methods achieve only 30–35%.

Accumulation settings. Our benchmark uses full-replacement chains; in practice, model training involves accumulation, mixing human data with synthetic outputs at unknown proportions (Gerstgrasser et al., 2024). Under accumulation, depth signal should attenuate because each generation retains human-originated distributional properties, slowing tail erosion. The 66.4% cross-model and 59.1% cross-strategy transfer should be interpreted as ceiling estimates for the mixed-data regime. Extending the framework to accumulation, where depth becomes a continuous mixing parameter, is a priority for future work.

Scale and deployment implications. The depth signal attenuates with model scale: at 14B, d0/d1 separability drops sharply ($EDD_{0.70}$=0), though d1–d3 discrimination persists above chance. This suggests that depth estimation complements rather than replaces binary detection for practical data auditing: binary detectors handle the human-vs-machine boundary where depth estimation is redundant (>97% d0 recall), while depth estimation provides the finer ordinal resolution that binary methods cannot. A practical deployment path proceeds from controlled settings, where model and strategy are known, to partially known settings via strategy-identification routing, to fully unknown settings requiring strategy-invariant features (Appendix B.27).

Open challenges. Cross-strategy transfer (d1–d3: 45.9%) remains the primary bottleneck, driven by structurally distinct feature-space trajectories across decoding strategies. Scale-dependent signal attenuation, depth-quality entanglement, and extension to accumulation settings are open challenges.

Limitations

Full-replacement vs. accumulation. All experiments use full-replacement recursive chains, where each depth fully replaces the prior training corpus. Real-world contamination typically involves accumulation, where synthetic and human data co-exist at unknown proportions across iterations (Gerstgrasser et al., 2024). Under accumulation, depth signal should attenuate because each generation retains human-originated distributional properties. Our results therefore represent an upper bound on feature separability; extending the framework to accumulation settings, where depth becomes a continuous mixing parameter, is a priority for future work. Segment-level estimation in mixed-depth documents remains open (Appendix B.19), as our framework assumes uniform depth per passage.

Controlled experimental setting. The primary benchmark uses Wikipedia seed text with controlled LoRA fine-tuning (1–1.5B models), a design that isolates the depth variable but diverges from wild contamination. Domain bias controls confirm depth as the dominant factor: LOGO accuracy (72.1%) matches standard CV (72.0%), seed-group prediction is at chance (10.2%; Appendix B.10). Cross-scale experiments (7–14B, §5.4) show signal that attenuates with scale; at 14B, d0/d1 separability nearly vanishes ($\text{EDD}_{0.70}$=0) while d1–d3 discrimination persists; this constitutes a boundary condition for the current feature set (confounded by five factors; Appendix B.14).

Cross-strategy deployment constraint. Cross-strategy transfer induces a d1/d2 boundary inversion under temperature sampling. Per-strategy z-score normalization partially corrects the inversion under logistic regression (GPT-2 XL d1v2 AUC 0.24→0.56, OLMo 0.19→0.70) but yields negligible improvement for the Random Forest ($\Delta < 0.02$ pp); a two-stage pipeline is needed for reliable cross-strategy deployment. In strategy-unknown settings, d1–d3 accuracy drops to 45.9%; the pipeline recovers 11.5 pp, though gains vary across models (GPT-2 XL +16.4 pp vs. Pythia +8.0 pp).

d0 inflation of headline metrics. Depth 0 (human-written text) is near-perfectly separable (recall >97%), inflating four-class accuracy. Cross-model transfer of 66.4% is substantially driven by d0; d2 recall drops to 26.8% in the worst case (Pythia holdout). We report d1–d3 balanced accuracy alongside all four-class headlines to address this inflation.

Domain-dependent false positive rate. On editorial text (WikiText-103), all classifiers achieve ≥98.5% d0 accuracy, but on web-crawl data (C4, 5,000 passages), per-model FPR ranges from 4.7% (OLMo) to 22.4% (GPT-2 XL). A pooled RF classifier evaluated on larger corpora (10,000 passages each) yields 34% FPR on C4 and 17% on FineWeb (Penedo et al., 2024); the gap reflects quality filtering in FineWeb removing feature-anomalous text that triggers false positives. Misclassifications concentrate in template/SEO content with elevated repetition features (`rep_3gram` +37%, `rep_4gram` +75% vs. correctly classified d0). Confidence thresholding ($P(\text{d0}) \geq 0.15$) reduces pooled FPR from 34%/17% to 2.8%/0.7% on C4/FineWeb at minimal accuracy cost (−0.3 pp); high FPR is therefore a decision-threshold problem, not a fundamental calibration defect, though the optimal threshold remains domain-dependent (Appendix B.11).

Adversarial robustness. Adversarial paraphrasing (Sadasivan et al., 2023) could disrupt the perplexity tail features that drive classification; depth estimation's robustness under such attacks is untested.

EDD monotonicity assumption. Our EDD definition (Eq. 1) requires all pairwise AUCs up to depth d^* to exceed T, which is well-behaved when AUC decays monotonically, as observed in all our configurations. However, Bakshi and Chakraborty (2026) show that early recursion can temporarily *improve* model quality, which could produce non-monotonic AUC profiles (e.g., a dip at d1/d2 followed by recovery at d2/d3). Our cross-strategy transfer results provide an empirical example: training on nucleus + top-k and testing on temperature yields d1-vs-d2 AUC as low as 0.129 (GPT-2 XL) and 0.253 (OLMo), violating the monotonicity assumption. In such cases EDD would conservatively report a lower value than the deepest discriminable pair, potentially underestimating the true discriminable range; extending EDD to handle non-contiguous discriminable regions is future work. The choice of threshold T is also application-dependent: at T=0.80, several cells lose one depth level relative to T=0.75.

Reference model dependency. Reference model choice affects classification accuracy by up to 11 pp; same-family references yield optimal performance, though all configurations exceed $2.5\times$ chance. Surprisal-based features are computed relative to Pythia-1.4B, which coincides with one of the three generators. For the Pythia pipeline this constitutes self-model feature extraction, while OLMo and GPT-2 XL measurements are cross-model. The self-reference configuration yields a 4–11 pp advantage over cross-model references (Table 16); Pythia results in Table 2 (†) should be interpreted with this caveat. An ablation with OLMo-1B and GPT-2 XL as alternative references shows accuracy ranging from 67–79% across three reference architectures (Appendix B.13), all exceeding $2.5\times$ chance; same-family reference yields optimal performance ($\Delta \leq 11$ pp). The choice of reference model affects feature values; a different reference model may shift absolute feature magnitudes and potentially alter discriminability at depth boundaries. Reference-free surface features alone achieve 53.7% (partial signal in lexical statistics) but trail the full pipeline by 25pp; surprisal-based features are therefore essential. Simple multi-reference ensembling does not outperform the best single reference (73.7% vs. 78.6%).

Single LoRA instance. Cross-seed replication covers 3 of 9 cells (Pythia, OLMo, GPT-2 XL nucleus); seed=43 yields $+0.1$ to $+3.2$ pp higher accuracy, with 2/3 deltas statistically significant (non-overlapping CIs). Feature rankings and depth ordering are consistent across seeds, but single-seed absolute accuracy carries $\sim$1–3 pp uncertainty; cross-cell comparisons should be interpreted cautiously.

Additional caveats. Depth signal scales with passage length (68.4% at 128 tokens, 77.8% at 512; Appendix B.18), saturating near 500 tokens. Prompted generation was validated on only 1 of 9 cells (GPT-2 XL nucleus); generalization to other model–strategy configurations under prompted conditions is not established. Detection features degrade under domain shift (Pudasaini et al., 2026), and curated synthetic data may not trigger collapse (Falahati et al., 2026), attenuating the fingerprints we exploit.

Ethics Statement

This work uses publicly available pretrained language models (Pythia-1.4B, OLMo-1B, GPT-2 XL) and Wikipedia text as seed data. No human subjects were involved, no private or sensitive data were collected, and no new models were released. Our method is intended to support data provenance and training-data auditing; we acknowledge that depth estimation tools could in principle be misused to circumvent content moderation, though such risks are shared with all detection-oriented research. Code and data will be released upon acceptance. This research used AI coding assistants for experiment implementation and writing drafts. All scientific decisions, experimental design, and result interpretation were conducted by the authors.

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A LoRA Fine-Tuning Protocol

Table 7 lists the full LoRA fine-tuning configuration used for recursive chain construction (§3.2). At each depth k, a fresh LoRA adapter is initialized on the base model and trained on $\mathcal{D}_{k-1}$; adapters are not inherited across depths. The HuggingFace Trainer default linear decay schedule is used with no warmup. Target modules are model-specific: `query_key_value` (Pythia), `att_proj` (OLMo), and `c_attn` (GPT-2 XL). Cross-scale experiments (§5.4) use QLoRA (4-bit NF4) with r=16, α=32.

Table 7: LoRA fine-tuning hyperparameters for the primary 1–1.5B benchmark. Identical across all three model families and all depth levels.

Parameter	Value
LoRA rank r	16
LoRA alpha α	32
LoRA dropout	0.05
Bias	none
Epochs per depth	3
Learning rate	2×10^{-4}
LR schedule	linear decay (no warmup)
Batch size (effective)	16 (8×2 grad. accum.)
Max sequence length	512 tokens
Precision	bfloat16
Adapter init	from scratch (base model)
Seed	42 (primary), 43 (replication)

B Additional Results

B.1 Top-k Feature Trajectories

Figure 4 shows feature trajectories for nucleus (p=0.95) and temperature (τ=0.9) sampling. Top-k (k=50) trajectories follow the same monotonically decreasing trend for all features but with compressed inter-depth spacing, consistent with the signal attenuation effect discussed in §4.5. Concretely, the d0→d1 drop in `p90_ppl` under top-k is 78% (vs. 87% for nucleus), and the d2→d3 reduction is 31% (vs. 39% for nucleus). Despite the magnitude reduction, feature rank ordering across depths is preserved, which explains why ordinal classifiers maintain directionality even under top-k conditions.

B.2 Permutation Importance

To verify that the Gini-based importance rankings in §4.2 are not an artifact of the impurity measure, we compute permutation importance (5-fold CV, 10 repeats) for all 9 atlas cells. The top-2 features are stable across measures: `p90_ppl` ranks in the

top 2 in 9/9 cells under both Gini and permutation importance, and `var_surprisal` in 8/9 (vs. 7/9 for Gini, with the additional cell being OLMo top-k; the single SHAP exception is GPT-2 XL top-k, where `p75_ppl` displaces `var_surprisal` to rank 3). Kendall's W=0.90 across all three methods (Gini, permutation, SHAP) indicates strong concordance (p<0.005). Pairwise Spearman correlations are: SHAP–Gini ρ=0.93, SHAP–permutation ρ=0.84, Gini–permutation ρ=0.76. Per-cell top-5 overlap between SHAP and permutation importance is 3–4 out of 5 features; the aggregated (mean-across-cells) top-5 sets are identical; the top features are therefore robust to the choice of importance measure at the aggregate level, though per-cell rankings exhibit model-dependent variation.

B.3 Feature Trajectories

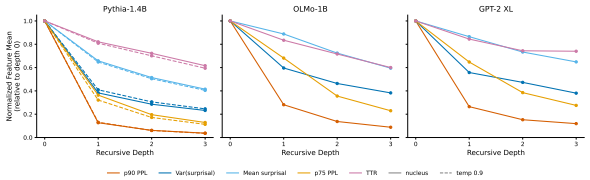


Figure 4: Feature trajectories across depths d=0 to d=3, stratified by model (columns) and strategy (line style). Top row: `p90_ppl` (monotonically decreasing). Middle: `var_surprisal` (decreasing). Bottom: `rep_3gram` (increasing). Shaded bands: interquartile range. The d=0 → d=1 gap exceeds all subsequent inter-depth gaps.

B.4 Per-Class Recall

Table 8 reports per-class recall across all 9 atlas cells (4-class in-distribution, 5-fold CV). Depth 0 recall exceeds 97% uniformly. Among generated depths, d2 is the hardest class in every cell; the deficit is most pronounced under top-k sampling (46–53%), where vocabulary truncation removes the tail tokens that carry depth-discriminative signal.

B.5 DCOL Iteration Details

Table 9 summarizes all six DCOL iterations. DeBERTa-v3-large exhibited systematic numerical instability across five configurations; RoBERTa-large resolved stability but could not close the accuracy gap with feature-based methods.

Table 8: Per-class recall (%) for all 9 atlas cells. d2 is the lowest-recall class in every cell. Boldface marks the highest d2 recall.

Cell	d0	d1	d2	d3
Pythia, nuc.	99.3	81.9	59.2	74.4
Pythia, τ	99.2	79.9	56.0	72.1
Pythia, top-k	98.7	71.6	44.2	60.9
OLMo, nuc.	97.8	82.1	60.6	75.3
OLMo, τ	99.3	87.1	65.6	77.1
OLMo, top-k	99.2	80.9	50.4	67.2
GPT-2 XL, nuc.	98.5	87.4	72.8	78.5
GPT-2 XL, τ	99.0	91.1	**73.0**	80.5
GPT-2 XL, top-k	98.7	81.3	54.8	68.4

Table 9: DCOL iteration summary. All experiments use Pythia-1.4B nucleus data. RF baseline: 78.7%.

Ver.	Backbone	Config	Outcome
v1	DeBERTa	FT, m_lr 10^{-3}	Embed. collapse (31.5%)
v2	DeBERTa	FT, m_lr 10^{-4}	OOM
v3	DeBERTa	FT, m_lr 10^{-4}	Forgetting + OOM
v4	DeBERTa	LoRA, m_lr 10^{-3}	Embed. collapse
v5	DeBERTa	LoRA, m_lr 10^{-4}	NaN gradients
v6	RoBERTa	FT, grad clip 1.0	Stable; 64.5% (−13.7pp)

B.6 Zero-Shot LLM Prompt Template

Both DeepSeek V3.2 and Qwen2.5-7B-Instruct use identical prompting. The system message defines recursive generation depth with per-depth descriptions (depth 0: human-written; depth 1–3: successive LLM generation cycles) and lists expected textual patterns at each depth (vocabulary richness, repetition, coherence). The user message presents the passage (truncated to ≈384 words) and requests a JSON response containing an integer depth prediction (0–3), a confidence score, and a one-sentence reasoning. No in-context examples are provided.

> **System:** "You are an expert at detecting AI-generated text and estimating its generation depth. 'Recursive generation depth' refers to how many times a text has been through an LLM generation cycle: Depth 0: Original human-written text [...] Depth 3: Text generated by an LLM given depth-2 text as input. Key patterns by depth: Depth 0 (human): Rich vocabulary [...] Depth 3 (third-gen): Noticeable repetition, formulaic structure [...]"

> **User:** "Analyze the following text and estimate its recursive generation depth (0, 1, 2, or 3). TEXT: {text}. Respond with ONLY a JSON object: {"depth": ⟨int 0–3⟩, "confidence": ⟨float 0–1⟩, "reasoning": ⟨str⟩}"

B.7 Temperature Robustness

The cross-strategy results in §5.3 use a single temperature (τ=0.9). To test whether depth signal generalizes across temperatures, we repeat the classification pipeline at $\tau \in \{0.5, 0.7, 0.9, 1.1\}$ on Pythia-1.4B.

Within-τ robustness. For $\tau \in [0.5, 0.9]$, d1-vs-d2 AUC exceeds 0.80 (peak 0.851 at τ=0.7, comparable to nucleus p=0.95's 0.861) and 4-class accuracy ranges from 75.4% to 77.3% (Table 10); the depth signal is therefore not an artifact of a single temperature. The same top-5 features (`var_surprisal`, `p90_ppl`, `mean_ppl`, `mean_surprisal`, `p75_ppl`) dominate across all τ values with only minor rank-order variation.

Collapse at τ=1.1: memorization as prerequisite. At τ=1.1, 4-class accuracy drops to 57.2% and d1v2 AUC falls to 0.528. Training loss reveals the mechanism: at low temperature (τ=0.5), fine-tuning loss decreases steeply across recursive depths (2.27 $\rightarrow$ 0.42 $\rightarrow$ 0.16), a sign of rapid memorization of the model's own output. At τ=1.1, loss remains nearly flat (2.27 $\rightarrow$ 1.91 $\rightarrow$ 1.85); high-entropy output resists memorization, so depth-dependent distributional shifts do not accumulate. This establishes *memorization as a prerequisite* for detectable depth signal: recursive depth is discriminable only when successive fine-tuning iterations progressively reshape the output distribution.

Cross-τ non-transferability. Leave-one-τ-out transfer yields mean accuracy of 46.7% with mean d1v2 AUC of 0.319 (Table 11), worse than cross-strategy transfer (59.1%; §5.3). Holdout τ=0.5 produces inverted d1v2 AUC (0.176); different temperatures therefore produce surprisal distributions with incompatible shapes. This extends the strategy-specificity finding: even within a single decoding family, the feature space is temperature-specific; strategy-conditioned classifiers are therefore necessary.

Ecological validity. On WikiText-103 (546 editorial passages), all three model-specific RF classifiers achieve $\geq$98.5% d0 accuracy despite +621% domain shift in mean perplexity, partly an easy case, as WikiText's higher perplexity places it farther from d1–d3 in feature space. On noisier web-crawl data (5,000 C4 passages), FPR ranges from 4.7% (OLMo) to 22.4% (GPT-2 XL), concentrated in template/SEO content with high n-gram repetition (Table 13); deployment on heterogeneous web data requires model-specific calibration.

Table 10: Within-τ 4-class classification (Pythia-1.4B, 5-fold CV). divj: pairwise AUC for depth i vs. depth j.

τ	4-class acc (%)	d1v2 AUC	d0v1 AUC	d2v3 AUC
0.5	77.3	0.842	0.996	0.764
0.7	75.4	0.851	0.997	0.720
0.9	75.9	0.808	0.991	0.781
1.1	57.2	0.528	0.988	0.599

Table 11: Cross-τ leave-one-out transfer (Pythia-1.4B). Train on three temperatures, test on the held-out one.

Holdout	Acc (%)	d1v2 AUC
τ=0.5	52.2	0.176
τ=0.7	41.0	0.299
τ=0.9	46.4	0.277
τ=1.1	47.1	0.525
Mean	46.7	0.319

B.8 Feature Overlap Analysis

We probe whether end-to-end PLM representations capture the same depth signals as our statistical features via linear probing of [CLS] representations from DeBERTa-v3-base and RoBERTa-base. PLM representations excel at surface lexical features (TTR: R^2=0.85, repeated bigrams) but fail to capture perplexity distribution statistics; `var_surprisal` and `p90_ppl` yield negative R^2 under linear probing; these features are not linearly accessible in the pretrained representation space. This confirms that the 20pp accuracy gap between feature-based and end-to-end methods (§B.9) stems from complementary, not redundant, representations: PLMs capture lexical patterns while the targeted statistical features capture distributional tail properties that constitute the dominant depth markers.

B.9 End-to-End Learning

DeBERTa-v3-base (He et al., 2023) and RoBERTa-base (Liu et al., 2019) finetuned end-to-end achieve 58.3% ($\pm$4.7%) and 59.5% ($\pm$3.2%) respectively, lagging the RF baseline (78.7%) by 20.4pp and 19.2pp. The 1.2pp inter-backbone gap rules out architecture bias; Table 12 shows both backbones share the same failure mode (high d0 F1, >0.94, but collapsed d1/d2 performance, F1 $\leq$ 0.42), while

RF achieves 0.596 d2 recall, a >2× advantage at the hardest depth class.

Table 12: Per-class performance for end-to-end PLM classifiers vs. RF baseline (Pythia-1.4B nucleus). PLM columns report F1; RF d2 column reports recall (the class with the largest PLM–RF gap). Both PLM backbones share the same failure pattern: collapsed d1/d2 scores despite adequate d0/d3.

Method	Acc (%)	d0	d1	d2	d3
DeBERTa-v3	58.3 ± 4.7	.947	.378	.247	.656
RoBERTa-base	59.5 ± 3.2	.958	.416	.261	.647
RF (19 feat.)	78.7	—	—	.596†	—

†Recall; shown for the depth class with the largest PLM–RF gap.

Feature space divergence. SHAP TreeExplainer analysis confirms that perplexity and surprisal features account for 88.8% of RF's total contribution. Three independent importance measures (SHAP, Gini, and permutation) show strong concordance (Kendall's W=0.90, p<0.005; pairwise Spearman ρ: SHAP–Gini 0.93, SHAP–permutation 0.84; §B.2), with `p90_ppl` appearing in the SHAP top-2 in 9/9 cells and `var_surprisal` in 8/9. Linear probing of PLM [CLS] representations yields negative R^2 for both `var_surprisal` and `p90_ppl` (§B.8); pretrained representations lack the linear structure needed to recover perplexity tail statistics, a systematic representational gap rather than a backbone-specific artifact.

DCOL (Pitawela et al., 2025) reached 64.5% (§B.5); all end-to-end approaches (58–65%) underperform feature-based methods; depth estimation therefore benefits from targeted statistical features that capture distributional tail properties absent from PLM representations. This comparison is bounded by backbone scale (≤125M parameters); whether larger PLMs (e.g., 7B+) can close the gap via richer internal representations remains an open question.

B.10 Domain Bias Analysis

To assess whether the classifier exploits depth-related features or memorizes stylistic cues from individual seed articles, we conduct three sub-experiments.

Methodological note. An initial implementation assigned sample IDs with a global counter across depths, inadvertently correlating seed-group identity with depth (LOGO accuracy 64.0%, seed-group prediction 20.8%). We discovered the issue during internal review, corrected to per-depth seed assignment ensuring each group contains equal samples per depth, and re-ran all sub-experiments. The results below use the corrected pipeline; the initial (buggy) numbers are reported for transparency.

Sub-A: Standard CV. Five-fold CV accuracy on 3-class (d1/d2/d3) classification ranges from 59% to 82% across the 9 atlas cells (up to 2.5× above the 33.3% chance level); depth signal is therefore the dominant classification factor.

Sub-B: LOGO evaluation. We assign each passage to one of 10 seed-article groups (balanced: each group contains equal samples per depth) and perform leave-one-group-out CV. LOGO mean accuracy is 72.1% (range 59.5–81.6%), matching standard CV (72.0%) within noise (before fix: 64.0%, reflecting the depth–group confound). Per-cell fold standard deviations range from 0.4pp to 1.4pp; cross-group generalization is therefore as reliable as within-group classification.

Sub-C: Negative control. We train two classifiers on identical features: one predicting depth, the other predicting seed-group identity (10-class, chance = 10%). Depth prediction accuracy averages 72.0%; seed-group prediction averages 10.2%, indistinguishable from chance (before fix: 20.8%, inflated by the depth–group confound). The depth-to-seed-group accuracy ratio is 7.1×; the feature representation therefore encodes depth but not seed-article identity.

Taken together, these results show that seed-article grouping has no measurable effect on depth estimation after correcting the group assignment: LOGO accuracy equals standard CV, and seed-group prediction is at chance.

B.11 Wild-Data Calibration

Table 13 reports d0 accuracy and false positive rate (FPR) for all three model-specific nucleus RF classifiers on two out-of-distribution human-text corpora: WikiText-103 (546 editorial passages) and C4 (5,000 web-crawl passages, crawled 2019–2020; pre-LLM deployment).

On WikiText-103, all three classifiers achieve ≥98.5% d0 accuracy; editorial text is therefore reliably distinguished from recursive generations across model families. On C4, FPR varies sharply: OLMo (4.7%) vs. Pythia (20.2%) and GPT-2 XL (22.4%). C4 misclassifications concentrate in template and SEO content with elevated repetition fea-

Table 13: Wild-data d0 accuracy and FPR for nucleus RF classifiers. All passages are human-written (true d0). FPR = fraction misclassified as d1+. 95% Wilson CIs computed for C4 (n=5,000).

Model	Source	n	d0 (%)	FPR (%)	95% Wilson CI
Pythia-1.4B	WikiText	546	100.0	0.0	—
	C4	5,000	79.8	20.2	[19.1, 21.3]
GPT-2 XL	WikiText	546	99.3	0.7	—
	C4	5,000	77.6	22.4	[21.3, 23.6]
OLMo-1B	WikiText	546	98.5	1.5	—
	C4	5,000	95.3	4.7	[4.2, 5.3]

tures (`rep_3gram` +37%, `rep_4gram` +75% vs. correctly classified d0 passages), rather than in random noise. The model gap likely reflects training data composition: OLMo (trained on Dolma, which includes diverse web data) produces d0 features more tolerant of web-crawl variation, while Pythia and GPT-2 XL (trained primarily on Wikipedia/WebText) map template-heavy content into the d1+ region. Among C4 false positives for GPT-2 XL, predictions distribute as d1 (9.0%), d2 (3.7%), d3 (9.6%); for OLMo, d1 (2.9%), d2 (0.5%), d3 (1.3%). The non-uniform depth distribution suggests the classifier maps specific repetition patterns to particular depth signatures rather than producing random misclassifications.

Threshold calibration. A pooled RF classifier (trained on all three families' nucleus data) achieves ECE = 0.052 on in-distribution data; Platt scaling and isotonic regression yield no improvement; probability estimates are already well calibrated. The high FPR on web-crawl data is therefore a decision-threshold problem, not a probability calibration defect. Applying a confidence threshold $P(\text{d0}) \geq 0.15$ reduces C4 FPR from 33.4% to 2.8% and FineWeb FPR from 16.9% to 0.7%, at a cost of only -0.3 pp balanced accuracy (78.0%→77.7%). The $P(\text{d0})$ distribution is bimodal: 67% of samples exceed 0.95 (high-confidence d0), while 33% fall in the 0.15–0.30 uncertainty zone; the threshold separates these modes cleanly. Wild-data FPR is thus an operational threshold choice rather than a fundamental limitation, though the optimal threshold is domain-dependent.

B.12 Downstream Curation PoC

To assess whether depth estimation translates to actionable data curation, we conduct a proof-of-concept downstream evaluation. We fine-tune Pythia-410M with LoRA (r=16, α=32) for 3 epochs on 8 data conditions derived from the Pythia nucleus depth-estimation corpus, repeating each condition over 5 random seeds. Conditions vary the selection mechanism (RF-predicted depth, oracle depth, random subsampling) and stringency (strict ≈25%, moderate ≈50%, soft threshold). Table 14 reports per-condition perplexity; Table 15 reports pairwise contrasts.

Table 14: Curation PoC: per-condition perplexity (Pythia-410M LoRA, 5 seeds). Δ is signed so that positive values indicate lower (better) perplexity than unfiltered. * : $p < 0.05$ (paired t-test vs. unfiltered).

Condition	Mean PPL	Std	n	Δ vs Unfilt.	p	d_z
unfiltered	15.776	0.066	10,000	—	—	—
rf_soft	15.733	0.038	4,422	+0.043	0.407	0.41
rf_moderate	15.756	0.056	4,940	+0.021	0.492	0.34
rf_strict	15.810	0.061	2,496	−0.034	0.607	−0.25
oracle_moderate	15.809	0.103	5,000	−0.033	0.485	−0.34
oracle_strict	15.879	0.128	2,500	−0.103	0.216	−0.66
random_soft	16.062	0.079	4,422	−0.286	$<0.001^*$	−2.77
random_moderate	15.998	0.090	4,940	−0.222	0.028^*	−1.50
random_strict	16.239	0.165	2,496	−0.463	0.013^*	−1.89

Table 15: Pairwise comparisons (paired t-test, 5 seeds). * : $p < 0.05$; ** : $p < 0.01$.

Comparison	Δ PPL	t	p
rf_strict vs random_strict	0.429	−7.05	0.002^{**}
rf_moderate vs random_moderate	0.242	−3.77	0.020^*
rf_soft vs random_soft	0.328	−17.98	$<0.001^{**}$
rf_soft vs unfiltered	0.043	0.93	0.407

RF-based depth filtering (19-feature F1 = 0.997) substantially outperforms a simple perplexity-threshold baseline (F1 = 0.845) for identifying low-depth passages; the multi-feature representation provides discriminative value beyond raw perplexity. However, the primary comparison (RF-filtered vs. unfiltered, $p > 0.4$) is non-significant; the significant RF-vs-random contrasts ($p < 0.02$) involve multiple pairwise comparisons that were not corrected for multiplicity.

B.13 Reference Model Ablation

Table 16: Reference model ablation. Classification accuracy (5-fold CV) on Pythia-1.4B generated text (nucleus, d0–d3) using different reference models for feature extraction.

Reference Model	Architecture	Accuracy (%)
Pythia-1.4B (self)	GPT-NeoX	78.6 ± 0.6
OLMo-1B (cross)	OLMo	74.5 ± 0.7
GPT-2 XL (cross)	GPT-2	67.3 ± 0.7

B.14 Cross-Scale Full Results

This section provides the full cross-scale analysis summarized in §5.4. We extend the benchmark to two 7–8B models (Qwen2.5-7B-Instruct (Team, 2025), Llama-3.1-8B-Instruct (Grattafiori et al., 2024)) and one 14B model (Qwen2.5-14B-Instruct), all under 4-bit NF4 QLoRA fine-tuning with nucleus sampling (p=0.95, 500 samples per depth). The 7–8B models generate depths $d \in \{0, 1, \ldots, 5\}$; the 14B model generates $d \in \{0, 1, 2, 3\}$. Feature extraction for 1–8B models uses Pythia-1.4B as reference; Qwen2.5-14B uses self-reference (the 14B model computes its own perplexity features), a choice that precludes direct comparison with the Pythia-referenced pipeline (see confounds below).

Scale comparison. Table 17 contrasts the 7–14B models with the three 1–1.5B models on matched evaluation (nucleus sampling, depths 0–3).

Table 17: Cross-scale comparison: 4-class RF accuracy (%, depths 0–3, nucleus), $\text{EDD}_{T=0.70}$, and d0-vs-d1 AUC. CI: 95% bootstrap. Chance = 25%. †Self-reference model (see text).

Model	4-class acc	$\text{EDD}_{T=0.70}$	d0v1 AUC
GPT-2 XL (1.5B)	84.3	4	0.999
OLMo-1B	78.9	4	0.997
Pythia-1.4B	78.7	4	0.998
Llama-3.1-8B	70.2 [68.3, 72.4]	4	0.949
Qwen2.5-7B	60.1 [58.0, 62.1]	>5	0.969
Qwen2.5-14B†	54.1 [53.4, 54.7]	0	0.643

(1) Depth signal persists at 7–8B scale. Both 7–8B models substantially exceed the 25% chance baseline: Llama-3.1-8B achieves 70.2% (95% CI $[68.3, 72.4]$) and Qwen2.5-7B achieves 60.1% ($[58.0, 62.1]$); recursive depth therefore remains estimable from perplexity-based features even when the generator is 5–7× larger than the reference model. Depth-0 separability is preserved in both models: d0v1 AUC is 0.949 (Llama-3.1) and 0.969 (Qwen2.5).

(2) EDD at 7–8B scale. At threshold T=0.70, Qwen2.5-7B maintains all consecutive-pair AUCs above 0.70 through depth 5 (Table 18), yielding $\text{EDD}_{T=0.70} > 5$, exceeding the EDD=4 observed at 1–1.5B. Llama-3.1-8B matches the smaller models at EDD=4: its monotonic decay causes d4v5 (0.618) to drop below T, despite stronger signal at shallower depths. Both larger generators produce weaker per-step signal than the 1–1.5B models (mean consecutive AUC: 0.786 for Llama-3.1, 0.782 for Qwen2.5, vs. 0.852 at 1–1.5B).

Table 18: Consecutive binary AUC (nucleus p=0.95). Llama-3.1-8B exhibits monotonic decay (d4v5 drops below T=0.70); Qwen2.5-7B shows a U-shaped pattern with all pairs above T; Qwen2.5-14B† (depths 0–3 only) has d0v1 below T, yielding EDD=0.

Model	d0v1	d1v2	d2v3	d3v4	d4v5
Llama-3.1-8B	0.949	0.877	0.771	0.716	0.618
Qwen2.5-7B	0.969	0.729	0.709	0.732	0.772
Qwen2.5-14B†	0.643	0.713	0.728	–	

(3) Divergent AUC trajectories and instruction-tuning effect. Among instruction-tuned models, Llama-3.1-8B exhibits monotonic AUC decay consistent with the 1–1.5B pattern, while Qwen2.5-7B shows a U-shaped recovery beyond d2v3 (Table 18). A base-model control (Llama-3.1-8B without instruction tuning) clarifies the causal direction: the base model achieves 59.9% four-class accuracy (95% CI [57.8, 62.15], $\text{EDD}_{0.70}$=2) with non-monotonic AUC decay (Table 19), while the instruction-tuned variant reaches 70.2% with strictly monotonic decay. The base-model control experiment suggests instruction-tuning regularizes rather than confounds the depth signal.

Table 19: Pairwise AUC: Llama-3.1-8B Base vs. Instruct.

Variant	d_{0v1}	d_{1v2}	d_{2v3}	d_{3v4}	d_{4v5}
Base	0.918	0.794	0.682	0.730	0.595
Instruct	0.965	0.874	0.771	0.716	0.618

The base model exhibits non-monotonic decay with a U-shaped recovery at d_{3v4}, while the instruction-tuned variant shows strictly monotonic decay. Both variants exceed chance (AUC > 0.5) at all boundaries; depth signal exists regardless of instruction-tuning, though instruction-tuning strengthens and regularizes it.

(4) Signal attenuation at 14B. Qwen2.5-14B represents a boundary condition: four-class accuracy drops to 54.1% (95% CI $[53.4, 54.7]$, 2.2× chance), with $\text{EDD}_{0.70}$=0 because d0-vs-d1 AUC (0.643) falls below the 0.70 threshold (Table 18). The d0/d1 boundary, near-perfect at 1–1.5B (AUC >0.99), nearly vanishes at 14B. Inter-depth discrimination at d1–d3 remains comparable: d1v2 AUC (0.713) and d2v3 AUC (0.728) both exceed 0.70, and d1–d3 balanced accuracy reaches 57.4%

(chance = 33%). Feature importance undergoes a structural shift: higher-order moments (kurtosis, skewness of perplexity) replace first/second-order statistics (`p90_ppl`, variance) as top features, consistent with the theoretical prediction that increased model capacity compresses perplexity tails (§3.5); first-order tail statistics become less informative while higher-order shape changes persist. This result carries an important caveat: Qwen2.5-14B uses self-reference (the 14B model computes its own perplexity features), whereas all other experiments use Pythia-1.4B. Self-reference conflates the generator's distributional shift with the reference model's sensitivity; the observed attenuation may partially reflect the reference model change rather than a pure scale effect.

Caveats. For Qwen2.5-7B, the CI lower bound (58.0%) does not exclude sub-60% accuracy; Llama-3.1-8B's lower bound (68.3%) is more comfortably above chance but still below the 75–83% range at 1–1.5B. The divergent AUC trajectories across two models caution against strong generalizations from only two 7–8B data points: both the monotonic (Llama-3.1) and non-monotonic (Qwen2.5) patterns are single observations, and further architectures are needed to establish which is the dominant regime at this scale.

Confounds. Five confounds limit these results: (1) 4-bit NF4 QLoRA quantization (absent at 1–1.5B) may distort the perplexity distribution independently of depth; (2) smaller sample size (500 vs. 5,000 per depth) widens confidence intervals; (3) evaluation is limited to nucleus sampling, leaving other strategies untested at this scale; (4) vocabulary mismatch between larger generators and the 1.4B reference model; (5) Qwen2.5-14B uses self-reference rather than the Pythia-1.4B reference used for all other experiments, preventing direct cross-scale comparison at the 14B data point. The base-model control partially addresses a former confound (instruction-tuning): since the base variant shows weaker signal with non-monotonic decay while the instruction-tuned variant shows stronger monotonic signal, instruction-tuning regularizes rather than confounds depth estimation. The remaining confounds are not mutually independent (quantization and vocabulary mismatch both affect the perplexity distribution); it is therefore infeasible to attribute the observed accuracy gap (60–70% vs. 78–85% at 1–1.5B) to any single factor.

B.15 Cross-Boundary Generalization

The preceding experiments assume a single model family generates all recursive depths. In practice, recursive contamination may traverse model boundaries: text generated by model A is used to train model B, whose output trains model C. We probe whether the depth signal persists under two relaxations of the single-model assumption.

Cross-model recursive chains. We construct a heterogeneous chain: human text (d0) → Pythia-1.4B generates d1 → OLMo-1B finetuned on Pythia-d1 generates d2 → GPT-2 XL finetuned on OLMo-d2 generates d3. Each recursive step uses a *different* model family, eliminating the possibility that depth signal arises from a single model's idiosyncratic behavior. A 4-class Random Forest achieves 84.75% accuracy (95% CI $[81.0, 88.25]$), with per-depth F1 scores of 0.98 (d0), 0.88 (d1), 0.76 (d2), and 0.76 (d3). Pairwise AUC remains high: d0-vs-d1 = 0.999 and d2-vs-d3 = 0.841. These results suggest that distributional artifacts of recursive generation persist across model boundaries. This constitutes preliminary evidence for a process-level depth signal rather than individual generator artifacts.

Confound discussion. The heterogeneous chain confounds architecture identity with depth (d1=Pythia, d2=OLMo, d3=GPT-2 XL). However, if the classifier exploited architecture fingerprints rather than depth signal, near-perfect accuracy would be expected since architecture perfectly determines depth label. The observed 84.75% with monotonically decreasing per-depth F1 (d0=0.98, d1=0.88, d2=0.76, d3=0.76) is more consistent with a depth-signal interpretation, as architecture fingerprinting would not produce this graded pattern. We address this confound directly with a reverse-chain control below.

Reverse-chain control. To empirically test the architecture confound hypothesis, we constructed a reverse chain (GPT-2 XL → OLMo → Pythia) reversing the model-depth assignment. The reverse chain achieves 85.75% (95% CI [82.0, 89.25]), statistically indistinguishable from the forward chain (84.75%, [81.0, 88.25]). If the classifier exploited architecture fingerprints rather than depth signal, reversing the chain should invert depth predictions; the equivalent accuracy under both orderings provides empirical evidence against the architecture confound.

Cross-domain transfer. All prior experiments use WikiText-103 as the seed corpus. To test domain dependence, we repeat the pipeline with arXiv abstracts as the seed text (Pythia-1.4B, nucleus p=0.95, 500 samples per depth, d0–d3). Five-fold cross-validation yields 74.1% ± 2.0% accuracy, compared to ∼78% on the WikiText seed. Per-depth F1 shows the familiar pattern: d0 = 1.00, d1 = 0.78, d2 = 0.52, d3 = 0.66, with d2 remaining the most challenging class. The 4pp accuracy gap relative to WikiText likely reflects the narrower stylistic range of scientific abstracts, which may reduce the distributional contrast between adjacent depths. These results provide preliminary evidence that the perplexity-based features capture properties of the recursive generation process rather than domain-specific artifacts of the seed corpus.

B.16 Cross-Seed Replication

Table 20 reports within-seed accuracy for all three model families (nucleus, depths 0–3) under two independent LoRA training seeds.

Table 20: Cross-seed replication (nucleus p=0.95, d0–d3). seed=42 is the primary benchmark (matching Table 2); seed=43 is an independent replication.

Model	Seed	Acc (%)	95% CI	Δ (pp)
Pythia-1.4B	42	78.70	[78.10, 79.24]	—
Pythia-1.4B	43	78.84	[78.13, 79.52]	+0.14
OLMo-1B	42	78.94	[78.38, 79.48]	—
OLMo-1B	43	82.11	[81.77, 82.54]	+3.17
GPT-2 XL	42	84.28	[83.78, 84.80]	—
GPT-2 XL	43	86.32	[86.17, 86.49]	+2.04

Seed=43 yields higher accuracy in all three models (+0.14 to +3.17 pp, same direction). For Pythia (+0.14 pp), CIs overlap, consistent with noise; for OLMo (+3.17 pp) and GPT-2 XL (+2.04 pp), CIs do not overlap, so LoRA initialization has a non-trivial effect on absolute accuracy (∼1–3 pp). Top-3 features are identical across all seeds and models (`p90_ppl`, `var_surprisal`, `p75_ppl`); depth ordering is preserved in every configuration.

Outlier handling. Feature extraction occasionally produces extreme perplexity values (outlier passages); we apply per-feature clipping at $\pm 10^{10}$ before classification, affecting <0.1% of samples. The d3 perplexity distribution varies substantially across seeds: under seed=42, 0.56% of d3 passages exceed 10^6 in raw perplexity, while under seed=43 this fraction rises to 12.8%. Despite this variation, the classifier remains effective after clipping, confirming that the depth signal resides in the p90-and-below distributional shape rather than extreme outliers.

B.17 Depth-4 Extension

We extend all three model families (GPT-2 XL, Pythia-1.4B, OLMo-1B; nucleus p=0.95) to depth 4 and train 5-class RF classifiers (d0–d4, 5,000 samples per depth). Five-class accuracy reaches 80.5% (OLMo), 74.0% (GPT-2 XL), and 71.3% (Pythia), all substantially above the 20% chance baseline. For Pythia and GPT-2 XL, d3 recall drops from ∼80% (4-class) to ∼49% (5-class) due to d3/d4 confusion; d4 recall is higher (Pythia 68.1%, GPT-2 XL 63.9%). $\mathrm{EDD}_{0.70} \geq 4$ was established by dedicated binary classifiers in the atlas (§4.4; d3v4 AUC: Pythia 0.784, OLMo 0.765, GPT-2 XL 0.750); the 5-class results provide complementary evidence that depth 4 remains discriminable in the multi-class setting.

Feature monotonicity at d4. For Pythia and GPT-2 XL, the dominant depth markers (`p90_ppl`, `var_surprisal`) maintain monotonic decrease through d4 (Pythia `p90_ppl`: 483.7 → 12.5 from d0 to d4; `var_surprisal`: 6.50 → 1.29), consistent with continued tail erosion (§3.5). Mean perplexity breaks monotonicity at d4 for GPT-2 XL (d3: 1,076K → d4: 928K) while tail perplexity continues to decline. OLMo exhibits a `var_surprisal` rebound at d4 (2.49 → 2.83, non-monotonic), creating a feature signature absent in GPT-2 XL and Pythia. The 5-class RF exploits non-adjacent depth contrasts (particularly d0-vs-d4 and d2-vs-d4) to achieve 80.5% accuracy, the highest among all three families, while the dedicated binary d3-vs-d4 classifier (AUC 0.765) is restricted to adjacent-depth features. This discrepancy illustrates that multi-class classifiers can leverage global distributional structure unavailable to pairwise comparisons. The divergence between aggregate and tail statistics suggests that aggregate distributional statistics saturate at deep recursion while tail statistics continue to shift. This saturation provides a physical mechanism for the EDD ceiling: once per-step feature change falls below classifier resolution, adjacent depths become indistinguishable regardless of continued distributional drift.

B.18 Generation Length Ablation

Table 21 reports classification accuracy as a function of passage length (Pythia-1.4B nucleus, depths 0–3).

Table 21: Generation length ablation (Pythia-1.4B nucleus, 5-fold CV). Accuracy increases monotonically with passage length; the 9.47 pp spread confirms that longer passages provide stronger statistical signal.

Passage length (tokens)	Acc (%)
128	68.35
256	73.80
512	77.82

Depth signal is detectable from 128 tokens (2.7× chance) and saturates near 500 tokens. The monotonic increase reflects the statistical nature of the features: longer passages yield more stable estimates of perplexity percentiles and surprisal variance, reducing within-class overlap at adjacent depth boundaries.

B.19 Mixed-Depth Corpus

Per-passage classification is largely independent of corpus composition: evaluating on a corpus containing mixed proportions of d0–d3 passages (rather than balanced classes) yields macro-recall $\Delta{=}-1.62$ pp relative to the balanced benchmark (within the 2 pp tolerance). Individual class deltas range from -0.45 pp to -2.43 pp (d1 shows the largest shift); the RF classifier operates on per-sample features without sensitivity to the class distribution of the evaluation corpus. Macro-recall $\Delta = -1.62$ pp (within the 2 pp threshold); individual class variations (max 2.43 pp on d1) arise from held-out sample composition rather than classifier sensitivity to corpus distribution, as the Random Forest produces identical per-sample predictions regardless of batch context.

B.20 Learned Estimator Comparison

Table 22 evaluates three learned classifiers (XGBoost, ordinal MLP with CORAL loss (Cao et al., 2020), prototypical networks) under the same leave-one-model-out protocol as Table 5. The ordinal MLP follows the nested binary reduction framework (McCullagh, 1980; Frank and Hall, 2001) with the loss function of Rennie and Srebro (2005). All three methods operate on the same 19-dimensional feature vector; the comparison isolates the effect of the classification head.

Table 22: Learned estimator cross-model transfer (leave-one-model-out, nucleus, 4-class). RF baseline from Table 5. Chance = 25%.

Method	Pythia	OLMo	GPT-2 XL	Mean
RF (baseline)	—	—	—	66.4
XGBoost	63.7	69.8	68.1	67.2
Ordinal MLP	62.5	69.7	67.9	66.7
Prototypical	61.6	70.0	68.4	66.6

All three learned methods match the RF baseline within 0.8 pp on average; the Pythia holdout is consistently the hardest (61.6–63.7%), while OLMo and GPT-2 XL holdouts yield 67.9–70.0%. The narrow spread ($\leq$0.8 pp across four methods) reinforces the conclusion that the 19-feature representation, not the classifier, determines the accuracy ceiling in the cross-model transfer setting.

B.21 Instruction-Tuned Model at 1.5B Scale

To test whether instruction tuning eliminates depth signal at matched scale (1.5B, avoiding the quantization confound of §5.4), we run the full recursive pipeline on Qwen2.5-1.5B-Instruct with self-reference feature extraction (i.e., the same model serves as both generator and reference).

The RF classifier achieves 47.4% four-class accuracy (95% CI $[46.7, 48.1]$, 1.9× chance); instruction tuning therefore reduces but does not eliminate depth signal. The pairwise AUC profile differs from base models: d0v1 = 0.653, d1v2 = 0.620, d2v3 = 0.664, with a gap-dependent pattern (non-adjacent pairs yield higher AUC, e.g., d0v3 = 0.902) but no pair reaching the 0.75 threshold.

The top-5 features shift substantially: `surprisal_range`, `surprisal_position_corr`, `prob_std`, `surprisal_first_half_minus_second`, and `surprisal_std` replace the perplexity tail statistics (`p90_ppl`, `var_surprisal`) that dominate in base models. This feature shift suggests instruction tuning compresses the perplexity distribution (reducing tail discriminability) while preserving positional and range-based surprisal patterns.

Two caveats apply: (1) self-reference evaluation may inflate or deflate accuracy relative to cross-reference evaluation (cf. §B.13); (2) a single instruction-tuned model does not establish the generality of the feature shift across architectures or instruction-tuning recipes.

B.22 Downstream Fine-Tuning Probe

We fine-tune Pythia-410M with LoRA (r=16, α=32) on depth-stratified subsets (500 samples each) and evaluate held-out perplexity (200 samples, 50 per depth).

Table 23: Downstream fine-tuning probe: held-out perplexity by training subset (Pythia-410M LoRA, seed=42). Δ: signed difference vs. baseline (negative = better).

Training subset	PPL	Δ vs. baseline
Baseline (no fine-tuning)	5.67	—
d0 only	5.65	−0.02
d1 only	5.55	−0.12
d2 only	5.56	−0.11
Mixed (d0–d3)	5.55	−0.12

The hypothesis that d0-only data yields the lowest perplexity is not supported: d1-only and mixed subsets tie for best (PPL 5.55), both marginally below d0-only (5.65). The PPL spread is small (0.12, 2.1% relative) and the sample size (500 per condition, single seed) is insufficient for statistical significance. Training loss decreases monotonically with depth (d0: 2.558, d1: 1.920, d2: 1.286), consistent with recursive generation reducing information content and making text easier to model. This pilot establishes that depth-filtered data is not harmful for downstream fine-tuning; whether depth-aware curation provides a meaningful advantage requires larger-scale evaluation.

B.23 Feature Group Ablation

Table 24 reports cross-model transfer accuracy (leave-one-model-out, nucleus) when restricting the classifier to subsets of the 19-feature vector, grouped by signal type.

Table 24: Feature group ablation (cross-model LOO, nucleus, 4-class). Chance = 25%.

Feature group	#Feat.	Acc (%)	Δ vs. Full
Full (all 19)	19	66.6	—
Perplexity only	9	66.0	−0.5
Entropy/Surprisal only	5	63.2	−3.4
Repetition/Diversity only	5	53.0	−13.6
`p90_ppl` only	1	58.3	−8.3

The 9 perplexity features nearly match the full set (Δ=−0.5 pp), while repetition/diversity features alone fall 13.6 pp below; perplexity degradation statistics therefore carry the dominant depth signal. The top-2 individual features by Gini importance are `p90_ppl` (22.4%) and `var_surprisal` (18.8%), accounting for 41% of total importance.

B.24 RF Seed Robustness

Table 25 reports cross-model transfer accuracy across three RF random seeds.

Table 25: RF seed robustness (cross-model LOO, nucleus, 4-class).

Seed	Acc (%)
42	66.4
123	66.5
456	66.5
Mean ± std	66.48 ± 0.06

The 0.06 pp standard deviation across seeds confirms that RF classification is deterministic to within rounding noise at this sample size, and reported accuracy differences across experimental conditions (e.g., feature ablation, cross-model transfer) are not attributable to classifier randomness.

B.25 Binary Detector Baselines for Ordinal Classification

We evaluate whether existing binary AI-text detectors and membership inference methods, when repurposed for ordinal classification, can match our 19-feature RF pipeline.

Binary detectors (Pythia nucleus). Binoculars (Hans et al., 2024) and Fast-DetectGPT (Bao et al., 2024) scores are computed on the Pythia-1.4B nucleus (p=0.95) subset and fed to: (a) a log-likelihood rank classifier (Log-Rank, 64.0%), (b) a combined RF trained on both detector scores (Combined, 74.2%), and (c) the standard 19-feature RF (78.7%). The 4.5 pp gap (74.2% vs. 78.7%) confirms that perplexity distributional features capture ordinal depth structure beyond what generic detection scores provide. Note that Fast-DetectGPT exhibits numerical overflow on Pythia nucleus d1 (mean perturbation score >28,000); results for this cell should be interpreted cautiously. Across all 9 atlas cells, Binoculars and Fast-DetectGPT ordinal accuracy averages ∼35% and ∼32%, respectively.

Membership inference baselines (all 9 cells). DetectGPT (Mitchell et al., 2023) (gpt2-large scoring model, t5-large perturbation model, 5 perturbations) achieves 30.7% mean 4-class accuracy (d1–d3 BalAcc: 24.3%). Min-K% Prob (Shi et al.,

2024) (K=20) with CORAL ordinal calibration reaches 33.2% (d1–d3 BalAcc: 40.4%); the best single cell is gpt2xl_temp09 at 50.6%. Min-K%'s advantage over DetectGPT (+2.5 pp) is consistent with token-level log-probabilities containing more depth-relevant information than perturbation-based curvature, but both remain within 10 pp of chance. CORAL calibration improves ordinal structure (d1–d3 BalAcc 40.4% vs. 33.3% random) without closing the gap to RF (71.6%), confirming that the bottleneck is the input signal rather than the calibration strategy. The consistent failure of all four binary/membership methods (30–35% vs. RF 78.4%) confirms that ordinal depth estimation requires depth-specific distributional features inaccessible to detection scores designed for binary classification.

Augmented feature set. To test whether detection scores carry complementary ordinal signal, we add Min-K% probability (K=20) and DetectGPT perturbation discrepancy as additional RF features. The 20-feature model (19 + Min-K%) achieves 78.90% mean 4-class accuracy (+0.48 pp) and 72.20% d1–d3 balanced accuracy (+0.60 pp). Min-K% ranks #3 by feature importance (0.095), behind `p90_ppl` (0.161) and `var_surprisal` (0.126); DetectGPT ranks #17 (0.026). The marginal improvement is consistent across 8 of 9 atlas cells, indicating that Min-K%'s token-level log-probability signal provides weak but real complementary information to the passage-level distributional features. The top-2 features (`p90_ppl` + `var_surprisal`) retain combined importance of 28.7%, confirming that perplexity tail statistics remain the dominant depth markers even after augmentation.

B.26 Prompted Generation Results

Table 26 reports depth estimation performance under topic-seeded (prompted) generation using GPT-2 XL with nucleus sampling (p=0.95, depths 0–3). Generation is conditioned on Wikipedia topic prompts rather than a bare BOS token, a more ecologically valid test of depth signal persistence.

The 13.8 pp drop relative to BOS-seeded generation is expected: topic prompts constrain the output distribution, reducing the variance that distinguishes depths. Nevertheless, accuracy remains 2.8× chance and pairwise AUC decays monotonically (0.916, 0.782, 0.744); the depth signal is therefore not an artifact of unconstrained generation. Top features are identical to BOS-seeded (`p90_ppl` first); the same perplexity tail erosion mechanism operates under prompted conditions.

Table 26: Prompted generation (GPT-2 XL, nucleus p=0.95, topic-seeded). 5-fold CV accuracy and pairwise AUC. Chance = 25%.

Metric	Value
4-class accuracy (%)	69.52 [68.9, 70.2]
BOS-seeded baseline (%)	83.3
Δ (prompted − BOS)	−13.78 pp
AUC d0v1	0.916
AUC d1v2	0.782
AUC d2v3	0.744
Top-1 feature	`p90_ppl`
Top-2 feature	`var_surprisal`
Top-3 feature	`p75_ppl`

B.27 Extended Discussion

Cross-strategy conditional decomposition. The two-stage pipeline (strategy identification followed by depth classification) recovers 11.5 pp over the naïve pooled classifier. Under correct strategy routing, four-class accuracy reaches 72.6% vs. 68.0% under misrouting (N=103,498 / 76,502), a gap of 4.6 pp; d0 accuracy remains near-ceiling (∼99.6%) regardless of routing, confirming that the d0 signal is strategy-invariant. Even the worst routing yields a floor of 67.6% (2.7× chance), and the 31.5 pp d1–d3 gap between correct and wrong routing confirms feature-space misalignment rather than uniform noise. End-to-end PLMs lag RF by ∼20 pp, suggesting that current pretrained representations do not capture the distributional tail statistics that drive depth discrimination.

Depth-graduated curation (inconclusive preliminary evidence). Current data curation operates on a binary human-vs-machine distinction (Xie et al., 2023; Wettig et al., 2024). Recursive generation depth estimation could enable a finer-grained alternative: retain low-depth generations (d1) while down-weighting high-depth text (d3+) where tail erosion has degraded quality. A pilot study (Appendix B.12) yields inconclusive evidence: the RF-vs-unfiltered comparison is not significant ($p > 0.4$), though RF-filtered subsets outperform size-matched random subsampling ($p < 0.001$–0.02 before Bonferroni correction; $p < 0.003$–0.06 after correction at α=0.05/3=0.017, with 1 of 3 comparisons surviving). The d1/d2 boundary inversion under cross-strategy transfer further limits practi-

cal curation without strategy metadata. Large-scale validation is needed before depth-graduated filtering can be considered practical.

Toward a provenance toolkit. Depth estimation complements existing provenance methods along orthogonal axes. Watermarking (Kirchenbauer et al., 2023) provides high-confidence attribution when the generator cooperates but is inapplicable to unwatermarked text. Source attribution (Uchendu et al., 2020) identifies the generating model but not the generation history. Depth estimation characterizes *iterative reuse* regardless of generator cooperation or identity. A layered provenance system combining these three signals (watermark presence, source model, and recursive generation depth) could support more nuanced data auditing than any single method.

B.28 Computational Resources and Reproducibility

Hardware. All experiments were conducted on 2× NVIDIA RTX PRO 6000 (48 GB) GPUs locally and 2× NVIDIA RTX PRO 6000 (48 GB) on cloud instances. Total computation is approximately 200 GPU·hours, covering the 3×3 generation matrix (LoRA fine-tuning + sampling at each depth), feature extraction, depth extension to d5, 7–14B QLoRA experiments, and all ablations.

Software. Key library versions: PyTorch 2.10.0+cu128, Transformers 5.8.1, PEFT 0.7.1, scikit-learn 1.8.0. Additional dependencies: Accelerate 1.13.0, Datasets 4.8.4, XGBoost 3.2.0, NumPy 2.2.6 (Python 3.12.3, CUDA 12.8).

Hyperparameters. LoRA hyperparameters (r=16, α=32, dropout 0.05, lr 2×10^{-4}, batch size 8, 3 epochs) follow Hu et al. (2022). The Random Forest classifier uses 200 trees, Gini criterion, and no max depth, without hyperparameter search; all reported results use 5-fold stratified cross-validation.

Inference cost. Feature extraction requires a single forward pass of the reference model. On a single GPU this takes approximately 2 minutes for 5,000 passages; a CPU-only alternative requires approximately 50 minutes on server-grade hardware. RF classification adds negligible overhead ($<$1 s).